Strange Matter Engine

Coupled map: CYP-target dynamical regimes

OpenADMET direct-inhibition validation trajectories | target-conditioned Graph Cellular Automaton

Quantity	Result
Trajectory observations	1,309
Distinct molecules	981
Distinct Murcko scaffold groups	906
CYP-label silhouette score	0.667
Multivariate permutation test	pseudo-F 7673.2; $p < 0.002$
Molecule-grouped target accuracy	0.998 +/- 0.002
Scaffold-grouped target accuracy	0.999 +/- 0.002

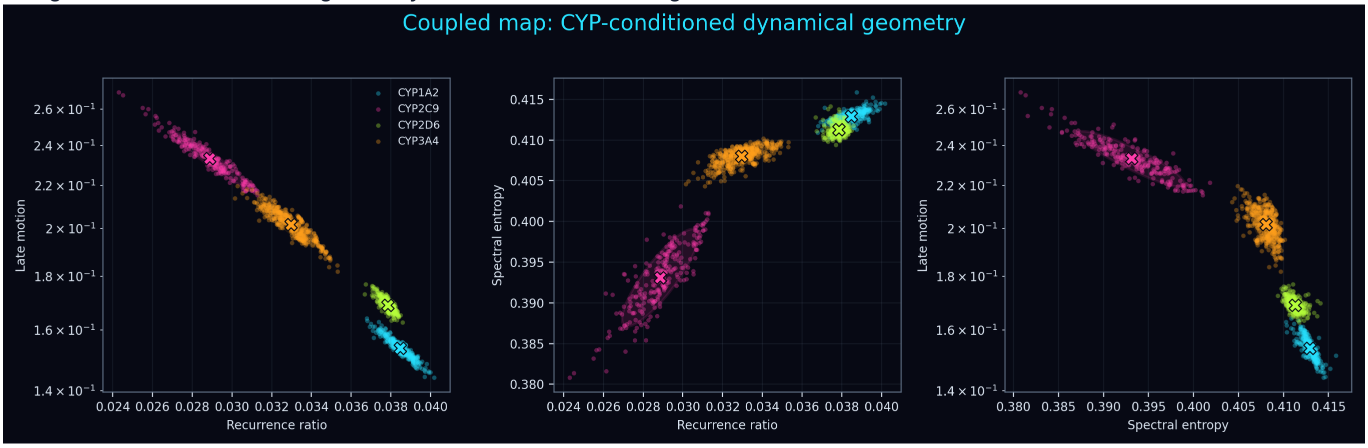
What the scores mean

A silhouette score near 1 indicates compact, well-separated CYP-labelled regions; a value near 0 indicates overlap. Balanced accuracy is measured against a 0.25 four-target chance level. Molecule-grouped folds prevent the same molecule entering both sides of a classification test. Scaffold-grouped folds additionally separate related chemical frameworks.

Target	Observations	Colour
CYP1A2	272	
CYP2C9	252	
CYP2D6	325	
CYP3A4	460	

Target-coloured dynamical geometry

Each point is one molecule-target validation observation. Translucent ellipses enclose approximately 80% of each target's bivariate Gaussian geometry; X markers denote target centroids.

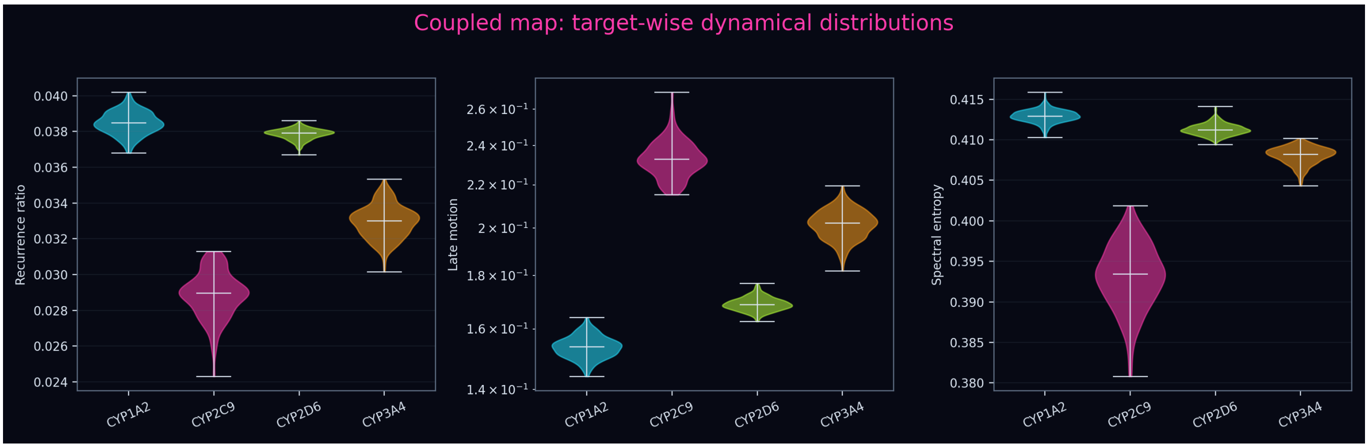


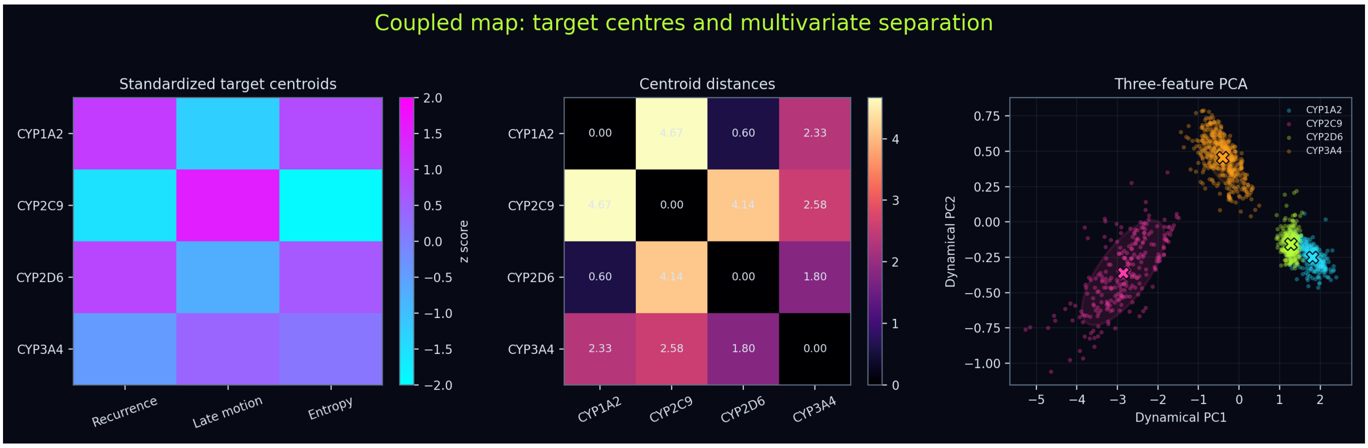
Target-specific effect sizes

Dynamical quantity	Variance associated with CYP target
Recurrence ratio	94.9%
Late motion	95.8%
Spectral entropy	93.2%

These eta-squared values quantify association, not causation. The CYP identity is supplied to the shared model as context, so strong separation means the learned dynamics use that context to create distinct regimes.

Distributions and target centroids

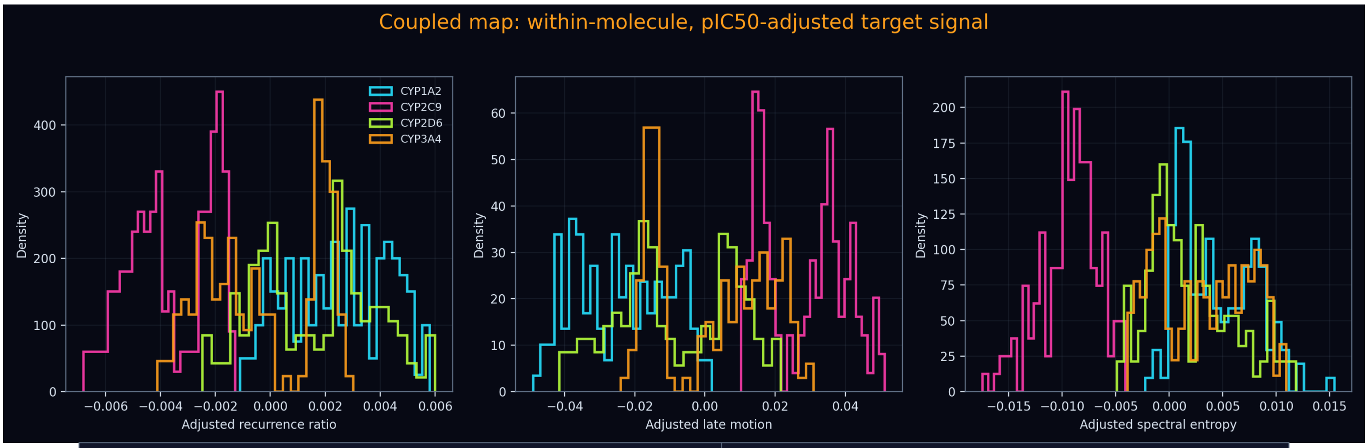




The centroid heatmap expresses every target mean in standard-deviation units. The distance matrix measures separation between those three-dimensional centres. PCA rotates the same three standardized variables into the two directions containing the most variance; it does not add information.

Does the target signal survive molecular and potency controls?

This stricter analysis retained 588 observations from 260 molecules having at least two CYP measurements. Each dynamical feature was centred within molecule, then its linear association with within-molecule pIC50 was removed. The remaining signal cannot be explained by a molecule's overall dynamical baseline or by a linear within-molecule potency difference.



Adjusted analysis	Result
CYP-label silhouette	0.155
Permutation pseudo-F	354.8
Permutation p value	0.002

Scientific interpretation

The analysis tests whether the trained Graph Cellular Automaton produces distinct trajectory-summary regimes for CYP1A2, CYP2C9, CYP2D6 and CYP3A4. Clear separation is evidence of target-conditioned internal dynamics. It does not by itself demonstrate four physical protein conformational states: the simulated states belong to the molecular graph model, and CYP identity is an explicit model input.

Recommended reading order

1. Inspect recurrence ratio versus late motion for visibly separated target regions. 2. Check spectral entropy to see whether the target also changes temporal complexity. 3. Compare target centroids and their distances. 4. Use scaffold-grouped accuracy to judge whether separation generalizes beyond related chemistry. 5. Use the within-molecule, pIC50-adjusted analysis to determine whether target structure remains after stringent controls.

Methods

The three trajectory summaries were standardized before multivariate calculations. Silhouette score used known CYP labels rather than an unsupervised cluster assignment. Multivariate pseudo-F significance used 499 label permutations. Logistic target classification used five stratified group folds and class-balanced multinomial regression. Murcko scaffolds were calculated from the standardized challenge SMILES. Confidence ellipses are descriptive Gaussian approximations. No model was retrained and no trajectory was recomputed.